2024-25 First Semester
MATH1053 Linear Algebra I

Assignment 7

Due Date: Due on 19/Nov/2024 (Tuesday), at the end of lecture.

- Write down your **CHN name** and **student ID**. Write neatly on **A4-sized** paper (*staple if necessary*).
- **Late submissions or answers without steps will not be graded.**

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1. If A is an $n \times n$ nonsingular matrix ($n > 1$), show that
$$\det(\text{adj}(A)) = (|A|)^{n-1}.$$

2. Prove that the matrix $A = I + \mathbf{u}\mathbf{u}^T$ is nonsingular where $\mathbf{u} \in \mathbb{R}^n$ is a column vector.

3. For real-valued square matrices A and B, prove that
$$\det \begin{bmatrix} A & B \\ B & A \end{bmatrix} = \det(A+B)\det(A-B).$$

4. Determine whether the following sets form subspaces of $\mathbb{R}^3$
   (a) $\{(x_1, x_2, x_3)^T | x_1 + x_3 = 1\}$
   (b) $\{(x_1, x_2, x_3)^T | x_1 = x_2 = x_3\}$

5. Determine whether the following are subspaces of P_4:
   (a) The set S_1 of polynomials in P_4 of even degree.
   (b) The set S_2 of all polynomials $p(x)$ in P_4 such that $p(0) = 0$.

6. Let U and V be subspaces of a vector space W. Define
$$U + V = \{\mathbf{z} \,|\mathbf{z} = \mathbf{u} + \mathbf{v}, \mathbf{u} \in U, \mathbf{v} \in V\}.$$
   Show that $U + V$ is a subspace of W.

7. Determine the null space of each of the following matrices:
$$\text{(a)} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \qquad \text{(b)} \begin{pmatrix} 1 & 2 & -3 & -2 \\ -2 & -4 & 6 & 3 \end{pmatrix}$$

8. Determine whether the following are spanning sets for $\mathbb{R}^2$. Justify your answers.
$$\text{(a)} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right\} \qquad \text{(b)} \left\{ \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ -4 \end{pmatrix} \right\}$$